

• RES.6-012 - Introduction to Probability - Lecture 4

• Lecture 4 - Counting

- Discrete uniform law
- Assume Ω consists of n equally likely elements.
- Assume A consists of k elements.

Then: $P(A) = \frac{\text{number of elements of } A}{\text{number of elements of } \Omega} = \frac{k}{n}$

→ Basic Counting principle

→ Applications

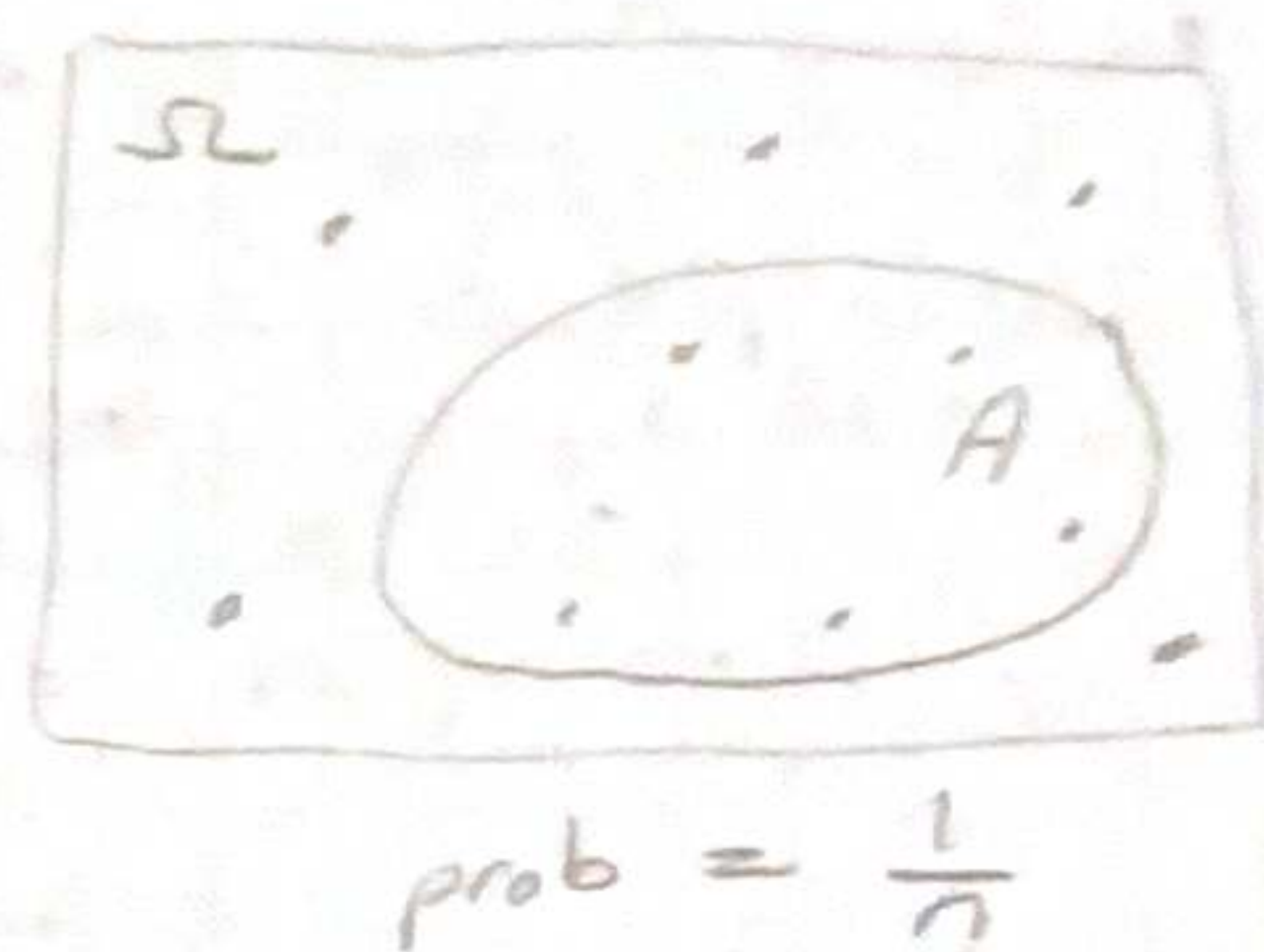
↳ permutations

↳ combinations

↳ partitions

↳ number of subsets

↳ binomial probabilities

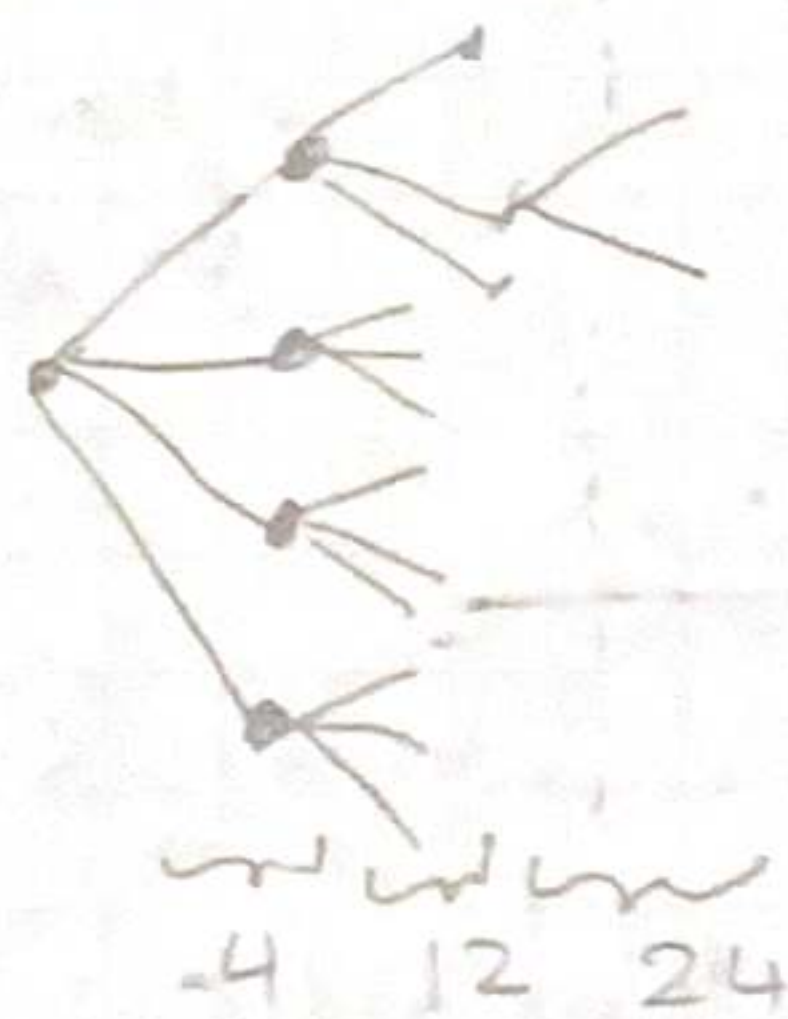


• The Counting Principle L4.2

→ Basic counting principle.

e.g. $\begin{matrix} 4 \text{ shirts} \\ 3 \text{ ties} \\ 2 \text{ jackets} \end{matrix} \left\{ \text{Number of possible attires?} \right.$

Options: $4 \cdot 3 \cdot 2 = \underline{24}$



• r stages

• n_i choices at stage i

↳ Number of choices is: $n_1 \cdot n_2 \cdot \dots \cdot n_r$

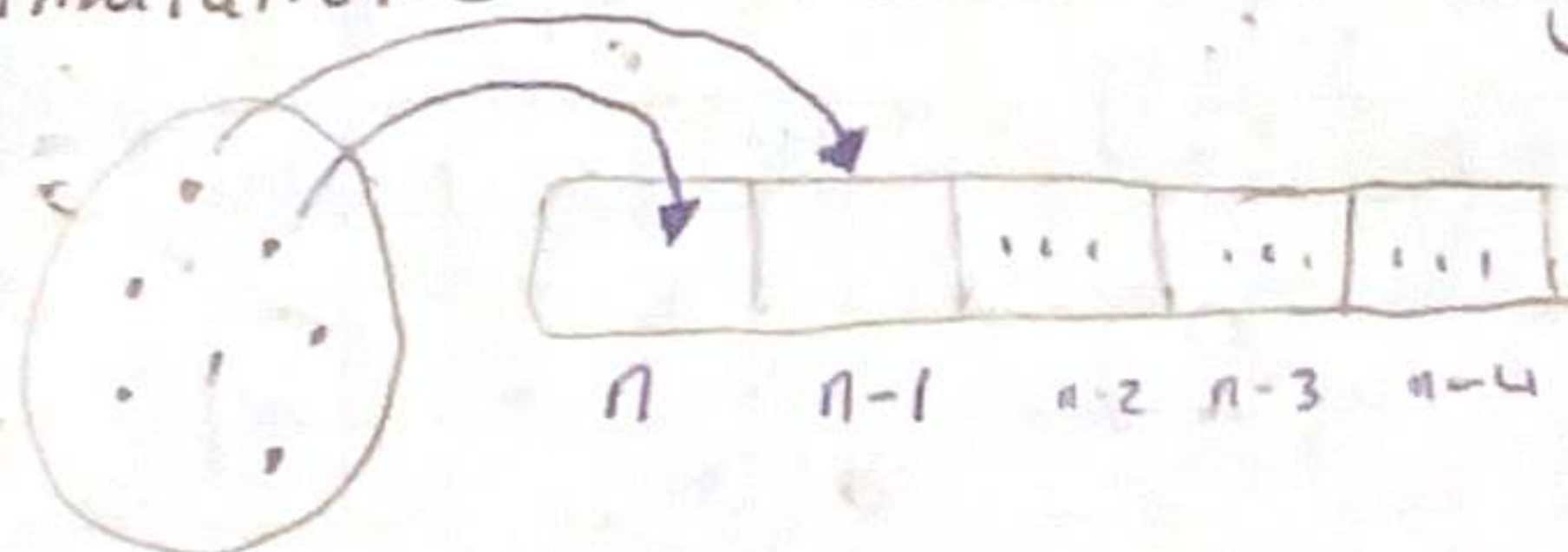
• Example 2:

↳ Number of license plates with 2 letters followed by 3 digits:

$26 \cdot 26 \cdot 10 \cdot 10 \cdot 10$

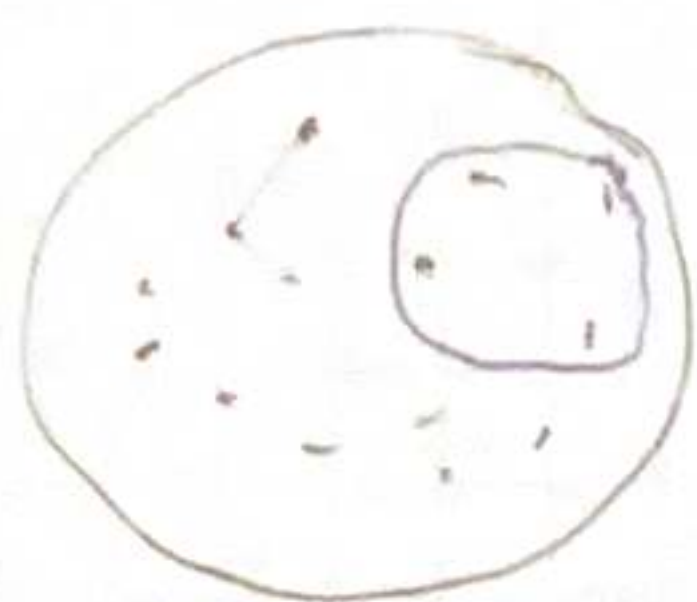
↳ What if repetition is prohibited: $26 \cdot 25 \cdot 10 \cdot 9 \cdot 8$

• Permutations: Number of ways of ordering n elements.



$$n(n-1)(n-2) \dots 1 = \boxed{n!}$$

- Number of subsets of $\{1, \dots, n\}$



$$2 \cdot 2 \cdots 2 = 2^n$$

case: $n=1$, $\{1\} \Rightarrow 2^n = 2^1 = 2$

$\hookrightarrow \{1\}$ or $\emptyset$ empty set.

• Die Roll Example L4.3

• Example:

- Find the probability that:
six rolls of a 6 (six-sided) die all give different numbers.
- Assume all outcomes equally likely.

$$\left\{ P(A) = \frac{\# \text{ in } A}{\# \text{ possible outcomes}} \right\}$$

Typical outcome

$$P(2, 3, 4, 3, 6, 2) = \frac{1}{6^6} \Rightarrow k^n$$

$$\Rightarrow P(\emptyset) = \frac{1}{k^n}$$

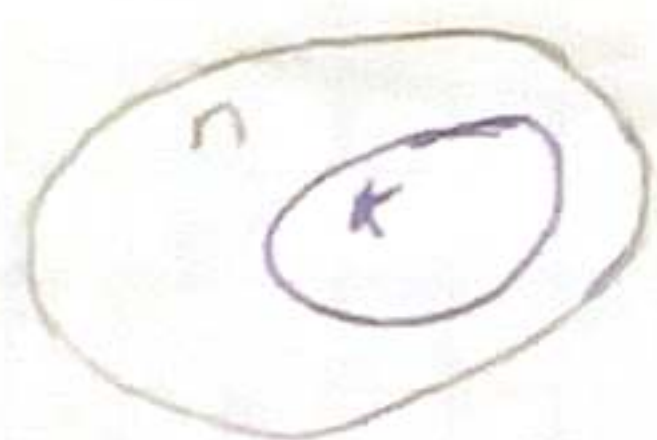
"element of A": $(2, 3, 4, 1, 6, 5) = 6!$

$\hookrightarrow$ How many way can we order the elements?

$\hookrightarrow$ Non-repeating $\rightarrow$ re-ordering.

• L4.4 Combinations

Start with set of n elements.



Definition

$\binom{n}{k}$: number of k -element subsets of a given n -element set.

- Two ways of constructing an ordered sequence of k distinct items:

- Choose the k items ~~one~~ one at a time

- Choose k items, then order them.

see $n(n-1)(n-2) \cdots (n-k+1) =$

$$\boxed{\frac{n!}{(n-k)!}}$$

OR

$\Rightarrow$

$$\boxed{= \frac{n!}{k!(n-k)!}}$$

$$\binom{n}{k} = \frac{n!}{k!(n-k)!} \quad (0! = 1)$$

$$\binom{n}{0} = \frac{n!}{0!n!} = 1 \Rightarrow \emptyset$$

• What about $\sum_{k=0}^n \binom{n}{k} = ?$

$$\sum_{k=0}^n \binom{n}{k} = \binom{n}{0} + \binom{n}{1} + \dots + \binom{n}{n} = \# \text{ of all subsets} = 2^n$$

• L4.5 Binomial Probabilities

Binomial coefficients $\binom{n}{k} \rightarrow$ Binomial probabilities.

$\hookrightarrow$ Related to probabilities used in Coin Tossing Model.

~~Example~~ : Coin Tosses

• Independent probabilities $\rightarrow$ can multiply together.

$$P(\text{Particular Sequence}) = p^{\# \text{heads}} (1-p)^{\# \text{tails}}$$

$$P(\text{particular } k\text{-head sequence}) = p^k (1-p)^{n-k}$$

$$P(k \text{ heads}) = p^k (1-p)^{n-k} \cdot (\# \text{ } k\text{-head sequences})$$

• L4.6 A Coin Toss Example

• Given 3 heads in 10 tosses.

• What is prob first two tosses ~~are~~ were heads?

- event A: the first 2 tosses were heads

- event B: 3 out of 10 tosses were heads.

Assumptions

$\hookrightarrow$ independence

$\hookrightarrow P(H) = p$

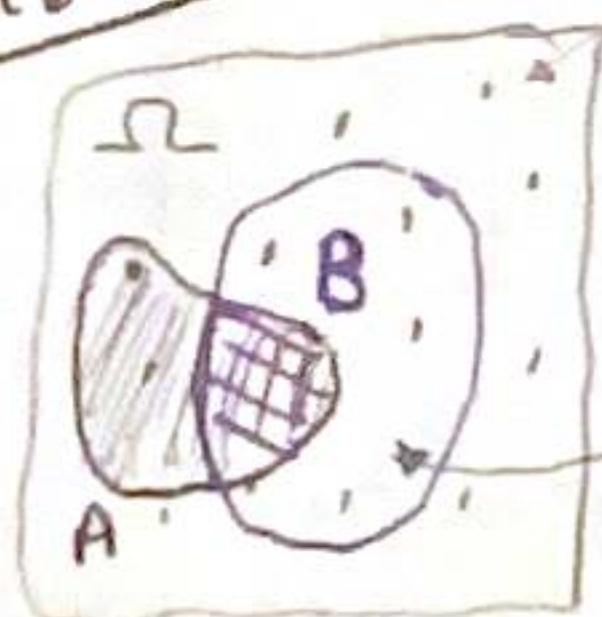
$$P(k \text{ heads}) = \binom{n}{k} p^k (1-p)^{n-k}$$

Method 1

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(H_1, H_2 \text{ and one } H \text{ in tosses } 3, \dots, 10)}{P(B)}$$

$$= \frac{p^2 \cdot \binom{8}{1} p^1 (1-p)^7}{\binom{10}{3} p^3 (1-p)^7} = \frac{\binom{8}{1}}{\binom{10}{3}} = \frac{8}{\binom{10}{3}}$$

Method 2



length 10 seq

3 head seq

$$P(3 \text{ head seq}) = p^3 (1-p)^7$$

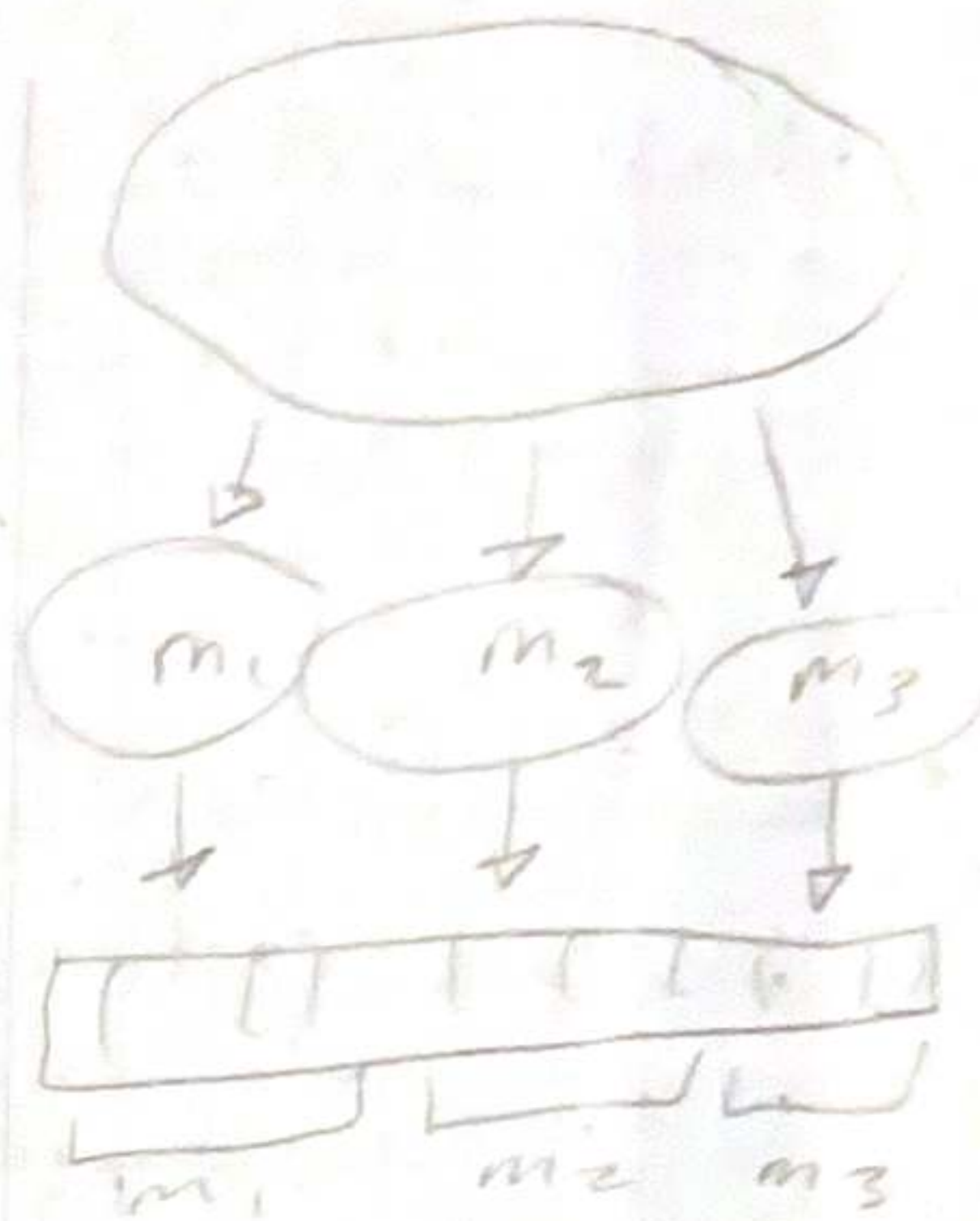
$$\frac{\# \text{ in } (A \cap B)}{\# \text{ in } B} = \frac{8}{\binom{10}{3}}$$

L4.7 Partitions

- $n \geq 1$ distinct items ; $r \geq 1$ persons
 give n_i items to person i
 - here $n_1, \dots, n_r$ are given nonnegative integers
 - with $n_1 + \dots + n_r = n$
- Ordering n items :
 - ↳ Deal n_i to each person i , and then order.

$$\text{number of partitions} = \frac{n!}{n_1! n_2! \dots n_r!}$$

(multinomial coefficient)



L4.8 Each Person Gets an Ace Example

Example : 52 card deck, dealt (fairly) to four players
 Find P (each player gets an ace)

- Outcome are : partition ; equally likely

- number of ~~partition~~ outcome :

$$\frac{52!}{13! 13! 13! 13!}$$

$$\frac{52}{4} = 13$$

- Constructing an outcome with one ace for each person :

- distribute the aces : 4, 3, 2, 1

- distribute remaining 48 cards : $\frac{48!}{12! 12! 12! 12!}$

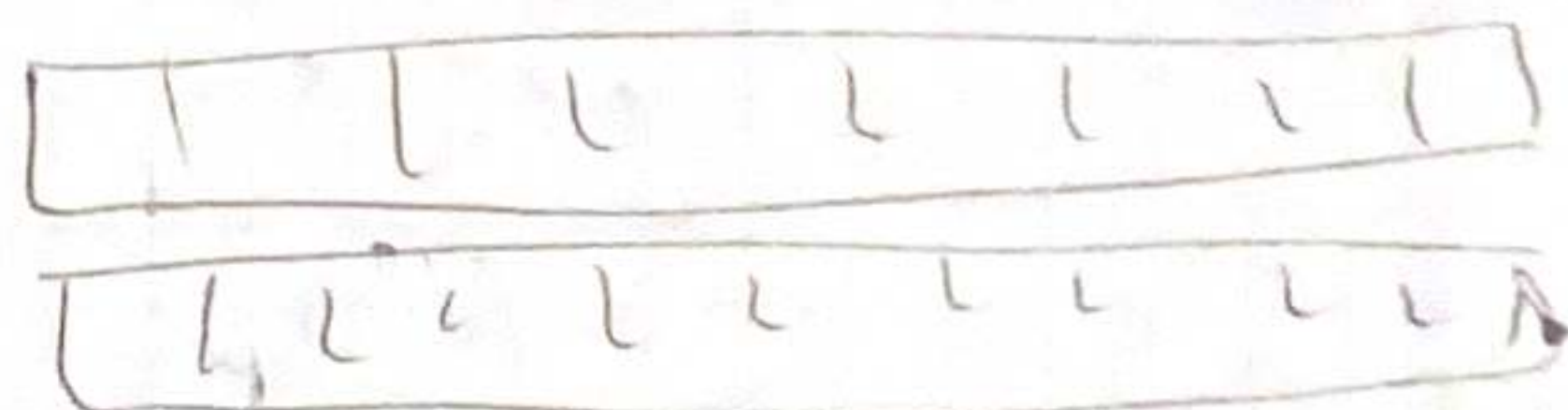
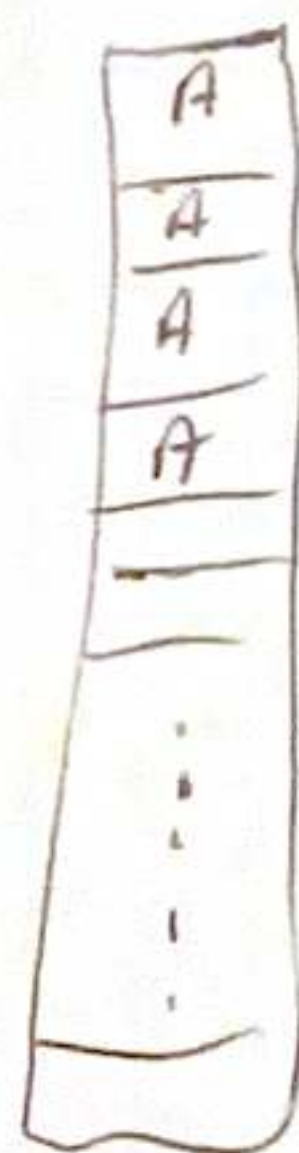
• Answer :

$$\frac{(4 \cdot 3 \cdot 2) \left(\frac{48!}{12! 12! 12! 12!} \right)}{\left(\frac{52!}{13! 13! 13! 13!} \right)}$$

Method 2 :

Stack the deck, aces on top

Players get 13 slots



$\frac{39}{51}$	$\frac{26}{50}$	$\frac{13}{49}$
$\underbrace{\hspace{1cm}}$	$\underbrace{\hspace{1cm}}$	$\underbrace{\hspace{1cm}}$
Prob A_2 in slot for diff person from A_1	A_3 not in persons where A_1 or A_2 are	A_4

$A_1 \rightarrow$ go any
 $A_2 \rightarrow$ anywhere but where A_1 is (51)

$$\Rightarrow P = 0.105$$

• L4.9 Multinomial Probabilities

- balls of different colours : $i = 1, \dots, r$ is p_i
- probability of picking a ball of colour $\underline{i}$ is p_i
- draw n balls, independently
- given nonnegative numbers n_i , with $n_1 + \dots + n_r = n$
- Find $P(n_1 \text{ balls of colour } \underline{1}, n_2 \text{ balls of colour } \underline{2}, \dots, n_r \text{ balls of colour } \underline{r})$
- Special case $r = 2$; colours : "heads", "tails"
↳ Similar to coin toss models.

$$n_1 \Leftrightarrow k \text{ heads} \quad p_1 = p$$

$$n_2 \Leftrightarrow n-k \text{ tails} \quad p_2 = 1-p$$

$$\binom{n}{k} p^k (1-p)^{n-k}$$

$$\hookrightarrow \frac{n!}{n_1! n_2!} \cdot p_1^{n_1} \cdot p_2^{n_2}$$

→ General Multinomial Probability:

$$P(\text{particular sequence of "type" } (n_1, n_2, \dots, n_r)) = p_1^{n_1} p_2^{n_2} p_3^{n_3} \dots p_r^{n_r}$$

sequence of type $(n_1, n_2, \dots, n_r) \Leftrightarrow$ partition of $\{1, \dots, n\}$ into subsets of sizes $n_1, n_2, \dots, n_r$

$$P(\text{get type } (n_1, n_2, \dots, n_r)) = \frac{n!}{n_1! n_2! \dots n_r!} \cdot p_1^{n_1} p_2^{n_2} p_3^{n_3} \dots p_r^{n_r}$$

• Introduction to Probability - Lecture 5 - Discrete Random Variables

• LS.1 Overview

- Random Variables?
- Discrete $\rightarrow$ takes values in a finite or countable set.
- Probability Mass Function (PMF)
- Examples of Random Variables
 - $\hookrightarrow$ Bernoulli
 - $\hookrightarrow$ Uniform
 - $\hookrightarrow$ Binomial
 - $\hookrightarrow$ Geometric
- Expectation (mean) $\rightarrow$ expected value rule
 $\rightarrow$ Linearity.

• LS.2 Definition of Random Variables

- Random Variable $\rightarrow$ numerical quantity that takes random values.
- Random Variable $\rightarrow$ a function that assigns a number to each outcome in a sample space.

Notation random variable X numerical value x

• LS.3 Probability Mass Functions (PMF)

- $\hookrightarrow$ Probability distribution of x .

• LS.4 Bernoulli & Indicator Random Variables

- Bernoulli Random variable $\rightarrow$ randomly either $[0; 1]$
- Indicator Random variable $\rightarrow I_A = 1$ iff A occurs.
 - $\hookrightarrow I_A = 1$ if
- $P_{I_A}(1) = P(I_A = 1) = P(A).$

• LS.5 Uniform Random Variables

$a \rightarrow b$ equally likely

• LS.6 Binomial RV

- $\hookrightarrow$ positive integer n ; $p \in [0, 1]$
- $\hookrightarrow$ Toss a coin.

• LS.7 Geometric RV

p : $0 < p \leq 1$

- $\hookrightarrow$ Infinitely many independent tosses of a coin.
- $\hookrightarrow$ models $\rightarrow$ waiting times, number of trials until success.

L5.8 Expectation

→ Mean of a random variable.

Average =

$$\text{Definition } E[X] = \sum_x x p_X(x)$$

↳ Average in large number of independent repetitions of experiment

L5.9 Elementary Properties of Expectations

$$\text{Definition } E[X] = \sum_x x p_X(x)$$

• If $X \geq 0$, then $E[X] \geq 0$ for all ω : $X(\omega) \geq 0$

• If $a \leq X \leq b$, then $a \leq E[X] \leq b$

• If c is a constant, $E[c] = c$

L5.10 Expected Value Rule

$$E[Y] = E[g(X)] = \sum_x g(x) p_X(x)$$

$$\text{e.g. } E[X^2] = \sum_x x^2 p_X(x) \neq (E[X])^2$$

L5.11) Linearity of Expectation

$$\text{Linear: } E[Y] = E[2X + 100] = 2E[X] + 100.$$

L5S11 Supplement: Functions

Functions

Discrete Random Variables Part II - Lecture 6 - Intro to Probability RES.6-012

L6.2 Variance

Variance $\rightarrow$ measure of the spread of a PMF (Probability Mass Function).

"new" $\rightarrow \mu$ • distance from mean: $X - \mu$.

$$E[X - \mu] = E[X] - \mu = \mu - \mu = 0$$

Definition - Variance: $\text{var}(X) = E[(X - \mu)^2] \geq 0$

$\hookrightarrow$ Standard Deviation: $\sigma_X = \sqrt{\text{var}(X)}$

Properties

• $\text{var}(aX + b) = a^2 \text{var}(X)$ $\rightarrow$ Move PMF by constant
 $\hookrightarrow$ Shifts distribution but does not affect its spread.

• Useful $\text{var}(X) = E[X^2] - (E[X])^2$

L6.3 Variance of Bernoulli & Uniform:

Bernoulli

$$X = \{0, 1\}$$

$$\text{var}(X) = \sum_i (x - E[X])^2 P_X(x) = p(1-p)$$

Uniform

$$\text{var}(X) = E[X^2] - (E[X])^2 = \frac{1}{12}(b-a)(b-a+1)$$

L6.4 Conditional PMFs & Expectations Given an Event

Conditional PMFs

$$P_{X|A}(x) = P(X=x|A) \quad \text{assume } P(A) > 0$$

$$\text{var}(X) = \frac{1}{12}(b-a)(b-a+1)$$

$$\text{Let } A = \{X=2\}$$

$E[X|A] =$ same as $E[X]$ within conditional model.

L6.5 Total Expectation Theorem

Conditionals are useful $\Rightarrow$ split complex models into simpler models.

$$P(B) = P(A_1)P(B|A_1) + \dots + P(A_n)P(B|A_n)$$

$$B = \{X = x\}$$

$$P_X(x) = P(A_1)P_{X|A_1}(x) + \dots + P(A_n)P_{X|A_n}(x)$$

$$E[X] = P(A_1)E[X|A_1] + \dots + P(A_n)E[X|A_n]$$

expectation of X = sum of proportional expectation A_1 to A_n
 $\hookrightarrow$ based off probability.

L6.6 Geometric PMF - Memorylessness & Expectation

$\rightarrow$ Conditional Geometric RV

Memorylessness $\rightarrow$ number of remaining coin tosses, conditional on Tails in the first toss, is Geometric, with parameter p .

$\Rightarrow$ Conditioned on $X > 1$, $X-1$ is geometric with parameter p .

$$P(3) = P(X-1 = 3 | X > 1) = P(T_2 T_3 H_4 | T_1) = P(T_2 T_3 | H_4) \\ \hookrightarrow \text{Coin toss example,} = (1-p)^2 p = P_X(3)$$

$$P_{X-1|X>1}(k) = P_X(k)$$

$$\Rightarrow P_{X-n|X>n}(k) = P_X(k)$$

• Mean of Geometric PMF

$$E[X] = \sum_{k=1}^{\infty} k P_X(k) = \sum_{k=1}^{\infty} k (1-p)^{k-1} p \\ = 1 + E[X-1]$$

$$E[X] = \frac{1}{p}$$

L6.7 Joint PMF & Expected Value Rule

Multiple RVs $\Rightarrow$ Joint PMFs

$X: p_x$
 $Y: p_y$ } knowing this does not tell you about $P(X=Y)$?

Joint PMF: $P_{X,Y}(x,y) = P(X=x \text{ and } Y=y)$

$$\hookrightarrow \sum_x \sum_y P_{X,Y}(x,y) = 1$$

$$P_{X,Y}(1,3) = \frac{2}{20}$$

$$P_X(4) = \frac{1}{20} + \frac{2}{20}$$

$$\therefore P(X=Y) = \frac{1}{20} + \frac{1}{20}$$

4	$\frac{1}{20}$	$\frac{2}{20}$	$\frac{4}{20}$	
3	$\frac{2}{20}$	$\frac{4}{20}$	$\frac{1}{20}$	$\frac{2}{20}$
2		$\frac{1}{20}$	$\frac{3}{20}$	$\frac{1}{20}$
1		$\frac{1}{20}$		
	1	2	3	4

$\hookrightarrow$ Scales to multi variables.

Expected Value Rule: $E[g(X,Y)] = \sum_x \sum_y g(x,y) P_{X,Y}(x,y)$

L6.8 Linearity of Expectations & Mean of Binomial

$$E[aX+b] = aE[X] + b$$

$\hookrightarrow$ So for: $E[X+Y] = E[X] + E[Y]$

$$E[X_1 + \dots + X_n] = E[X_1] + \dots + E[X_n]$$

Mean of Binomial

X : binomial with parameters n, p
- num successes in n independent trials

$$E[X] = \sum_{k=0}^n k \binom{n}{k} p^k (1-p)^{n-k} \} \rightarrow \therefore E[X] = np$$

$P_X(k)$

$$\{X = X_1 + \dots + X_n\}$$

$X_i = 1 \rightarrow$ success
 $X_i = 0 \rightarrow$ fail

} indicator variable $\Rightarrow E[X] = E[X_1] + \dots + E[X_n] = np$

(Lecture 4)

• Introduction to Probability Textbook Notes

• Textbook - Section 1.6 pg 44 - Counting

• The Counting Principle

↳ Divide and conquer approach.

Counting answers a simple question: when all outcomes are equally likely, how many outcomes make up my event?

$$P(A) = \frac{\text{number of outcomes in } A}{\text{total number of outcomes in } \Omega}$$

→ Counting Principle is foundational

↳ If a process has $\square$ sequential stages $n_1, n_2, n_3, \dots, n_r$ choices at each stage, then total outcomes: $\{ n_1 \cdot n_2 \cdot n_3 \cdot \dots \cdot n_r \}$

• Permutations

↳ Order matters & no repetition

→ Choosing and arranging k objects from n .

$$\frac{n!}{(n-k)!} = n(n-1)(n-2) \cdot \dots \cdot (n-k+1)$$

e.g. How many 4-letter sequences from the alphabet? $\Rightarrow 26 \cdot 25 \cdot 24 \cdot 23 = 358,800$

Q → what elements are in the group, and in what sequence?

• Combinations

↳ ~~Order~~ Order does not matter; no repetition.

↳ Choosing k objects from n with no regard to arrangement.

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

⊗

↳ Every combination corresponds to $k!$ permutations (the orderings of that group).

Q → what elements are in the group?

• Partitions

↳ Splitting n objects into r groups of sizes $n_1, n_2, \dots, n_r$

$$\frac{n!}{n_1! \cdot n_2! \cdot \dots \cdot n_r!}$$

↳ This is the multinomial coefficient — a generalisation of combinations.

↳ How many arrangements of TATTOO? $\frac{6!}{3! \cdot 2! \cdot 1!} = 60$

• Problem Solving Decision Matrix

Question to ask	Tool
Does order matter? Yes; no repetition	Permutation
Does order matter? No; no repetition	Combination
Splitting into labelled groups?	Partition / Multinomial
Sequential Stages, independent choices	Counting principle directly

• Additional ideas

① Sampling with vs. without replacement

↳ Without replacement → (the batch problem) pool shrinks with each draw

↳ Combinations or sequential multiplication

↳ With replacement → pool stays consistent → each stage has n choices

↳ pure counting principle.

② Overcounting is a strategy

↳ Sometimes it is easier to count a larger set and then divide out the duplicates

↳ Exactly how combinations are derived from permutations.

↳ See pattern repeatedly:

$$\text{combinations} = \frac{\text{permutations}}{k!} \quad \leftarrow \text{Divide out overcounting}$$

(Lecture 5 + 6)

- Introduction to Probability Textbook Notes
Section 2.1 → **2.5** - Discrete Random Variables

• Section 2.1 Basic Concepts

- A random variable is a real-valued function on the sample space.
- Every outcome gets mapped to a number.
 - Discrete - takes a finite or countably infinite set of values (integers, counts)
 - Continuous - takes values from an uncountable set (any real number in an interval).

↳ Most ideas from Chapter 1's event-based probability fundamentals remain relevant to random variables. Difference is applying familiar principles to numbers as opposed to abstract events.

• Section 2.2 Probability Mass Functions

↳ PMF fully characterises discrete random variables.

↳ For every possible value x :

$$P_X(x) = P(X=x)$$

→ two rules must hold:

$$P_X(x) \geq 0 \quad \text{and} \quad \sum_x P_X(x) = 1$$

↳ these rules follow directly from the axioms of probability.

• Distributions :

→ Bernoulli - single trial; success or failure $P_X(k) = \begin{cases} p & k=1 \\ 1-p & k=0 \end{cases}$

→ Binomial - number of successes in n independent trials.

$$P_X(k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad k = 0, 1, \dots, n.$$

→ Geometric - number of trials until first success.

$$P_X(k) = (1-p)^{k-1} \cdot p, \quad k = 1, 2, \dots$$

→ Poisson - count of rare events, parameter $\lambda > 0$

$$P_X(k) = e^{-\lambda} \cdot \frac{\lambda^k}{k!}, \quad k = 0, 1, 2, \dots$$

↳ Poisson is the Binomial's limit:

↳ limit as $n \rightarrow \infty$, $p \rightarrow 0$, with $np = \lambda$ held fixed.

↳ When n is large and p is small, use Poisson

$\lambda \rightarrow$ rate $\rightarrow$ average # events expected in given interval

$k \rightarrow$ actual count of events you want prob of.

• Section 2.3 Functions of Random Variables

→ If $Y = g(X)$, then Y is itself a random variable.

→ PMF:

$$Pr(y) = \sum_{\{x: g(x)=y\}} P_X(x)$$

↳ Collect all x values that map to the same y , add their probabilities.

↳ Example:

X uniform on $\{-4, \dots, 4\}$, $Y = |X|$.

Two values of X (e.g. ± 1) map to same $y=1$, so

$$Pr(1) = P_X(1) + P_X(-1) = 2/9$$

$\begin{matrix} -4 & \rightarrow & 0 \\ +4 & \rightarrow & 0 \end{matrix} \quad \begin{matrix} 4 \\ 4 \end{matrix} \rightarrow 1$

• Section 2.4 Expectation, Mean, and Variance

→ Expectation is the probability-weighted average:

$$E[X] = \sum_x x \cdot P_X(x)$$

→ Expected Value Rule

↳ For any function $g(X)$, compute the expected value directly from X 's PMF.

$$E[g(X)] = \sum_x g(x) \cdot P_X(x)$$

→ Variance - measures spread around the mean.

$$\text{var}(x) = E[(x - E[x])^2]$$

OR

$$\text{var}(x) = E[x^2] - (E[x])^2$$

↳ Calculate $E[x^2]$ using the Expected Value Rule with

↳ $g(x) = x^2$

↳ then subtract the square of the mean.

→ Linear transformations $Y = aX + b$

↳ $E[Y] = aE[X] + b$

↳ $\text{var}(Y) = a^2 \cdot \text{var}(X)$

↳ Note: constants (e.g. b) → shift mean $E[X]$.

↳ have no effect on variance $\text{var}(X)$.

↳ shifts the entire distribution rigidly

↳ distribution is translated

↳ spread does not change, hence $\text{var}(X)$.

↳ Note #2: Unless g is linear, $E[g(X)] \neq g(E[X])$

↳ average speed not a reciprocal of average travel time.

→ Distributions:

<u>type</u>	<u>mean</u>	<u>variance</u>
Bernoulli (p)	p	$p(1-p)$
Binomial (n, p)	np	$np(1-p)$
Geometric (p)	$1/p$	$(1-p)/p^2$
Poisson (λ)	λ	λ
Uniform $[a, b]$	$(a+b)/2$	$\frac{(b-a)(b-a+1)}{12}$

• Section 2.5 Joint PMFs of Multiple Random Variables

- When two Random Variables X and Y exist in the same experiment
↳ Required to track their joint behaviour

$$P_{X,Y}(x, y) = P(X = x, Y = y)$$

- ↳ Recovering individual distributions → Marginalisation:

$$P_X(x) = \sum_y P_{X,Y}(x, y) \quad P_Y(y) = \sum_x P_{X,Y}(x, y)$$

- ↳ Sum out the variable not interested in.

- ↳ P_X and P_Y → called Marginal PMFs

- ↳ Joint contains more information than either Marginal alone.

↳ Expected Value Rule

- ↳ extends naturally

$$E[g(X, Y)] = \sum_x \sum_y g(x, y) \cdot P_{X,Y}(x, y)$$

- ↳ Linearity of expectation

$$E[aX + bY + c] = aE[X] + bE[Y] + c$$

- ↳ holds even when X and Y are dependent.

- ↳ Variance does not have this property.

- ⇒ Pattern: Decompose a complicated RV into a sum of simple indicator RVs, apply linearity, compute n easy expectations instead of one hard one

- For variance: $\text{var}(X+Y) = \text{var}(X) + \text{var}(Y)$ only if X, Y independent

- ↳ If dependent: $\text{var}(X+Y) = \text{var}(X) + \text{var}(Y) + 2\text{cov}(X, Y)$

covariance.

Defining Probability Mass Functions: Example

Experiment: Roll two fair six-sided dice.

Let X = the sum of the two rolls.

Step 1: Define Sample Space

Each outcome is a pair (d_1, d_2) where $d_1, d_2 \in \{1, 2, 3, 4, 5, 6\}$

Total Outcomes = $6 \cdot 6 = 36$ (all equally likely)

Step 2: Identify possible values of X

Minimum sum is $1+1=2$
Maximum sum is $6+6=12$ } So $X \in \{2, 3, \dots, 12\}$.

Step 3: Count outcomes for each value of X .

For each value x , count how many pairs (d_1, d_2) sum to x .

x	Pairs	$P_X(x)$
2	(1,1)	$1/36$
3	(1,2) ; (2,1)	$2/36$
4	(1,3) ; (2,2) ; (3,1)	$3/36$
5	(1,4) ; (2,3) ; (3,2) ; (4,1)	$4/36$
6	(1,5) ; (2,4) ; (3,3) ; (4,2) ; (5,1)	$5/36$
7	(1,6) ; (2,5) ; (3,4) ; (4,3) ; (5,2) ; (6,1)	$6/36$
8	(2,6) ; (3,5) ; (4,4) ; (5,3) ; (6,2)	$5/36$
9	(3,6) ; (4,5) ; (5,4) ; (6,3)	$4/36$
10	(4,6) ; (5,5) ; (6,4)	$3/36$
11	(5,6) ; (6,5)	$2/36$
12	(6,6)	$1/36$

Step 4: Verify PMF sums to 1

The counts are 1, 2, 3, 4, 5, 6, 5, 4, 3, 2, 1 - summing to 36.

$\hookrightarrow \frac{1}{36} + \frac{2}{36} + \dots + \frac{1}{36} \Rightarrow 1$. $\hookrightarrow$ divide by 36